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AI-Based Energy-Saving for Fog Computing-Empowered Data Centers

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***Abstract*— With the rapid development of cloud computing services and big data applications, the number of data centers is proliferating, and with it, the problem of energy consumption in data centers is becoming more and more serious. Data center energy-saving has received more and more attention as a way to reduce carbon emissions and power costs. The main energy consumption of data centers lies in IT equipment energy consumption and end air conditioning energy consumption. In this paper, we propose a data center energy-saving application system based on fog computing architecture to reduce air conditioning energy consumption, and thus reduce data center energy consumption. Specifically, the intelligent module is placed in the fog node to take advantage of the low latency, proximal computing, and proximal storage of fog computing to shorten the network call link and improve the stability of acquiring energy-saving policies and the frequency of energy-saving regulation, thus solving the disadvantages of high latency and instability in the energy-saving approach of cloud computing architecture. The AI technology is used in the intelligent module to generate energy-saving strategies and remotely regulate the end air conditioners to achieve better energy-saving effects. This solves the shortcomings of the traditional manual regulation based on expert experience with low adjustment frequency and serious loss of cooling capacity of the terminal air conditioner. According to the experimental results, statistics show that compared with the traditional manual regulation based on expert experience, the data center energy-saving application based on fog computing can operate safely and efficiently, and reduce the PUE to 1.04. Compared with the AI energy-saving strategy based on cloud computing, the AI energy-saving strategy based on fog computing generates strategies faster and with lower latency, and the speed is increased by 29.84%.**

Keywords-data center energy-saving;fog computing; AI energy-saving; energy-saving system

I. INTRODUCTION

In recent years, great changes have taken place in the global climate. Climate warming is a general trend. Glacier melting also happens from time to time, posing a great threat to the normal life of coastal countries and some marine countries and regions. Energy conservation and emission reduction have also attracted the attention of many countries. With the popularization of intelligent applications, the demand for computing power is growing[1]. Data Center is an important infrastructure to provide computing power, and it also accounts for a large proportion of power consumption [2].According to statistics, Google's power consumption reaches 26 million watts per month in order to maintain the data center's normal operation and service[3].

To reduce the power consumption of the data center, many scholars have carried out relevant work in different directions[4]. At present, these works are mainly divided into two categories:

(1) Optimization of resources

The resources in the data center mainly include CPU, storage, switch, network [5], etc. In papers[6-7], the power sensing scheduling algorithm is proposed, which uses the dynamic voltage frequency scaling (DVFS) technology to make the processor work at different voltages and frequencies according to different loads, and select the appropriate power supply voltage and processing element frequency to minimize energy consumption without violating the SLA.

(2) Optimization of a cooling system in Data Center

The data center mainly has two kinds of cooling systems: air-cooling air conditioning and water-cooling air conditioning [8-9].The air-cooling air conditioning system inputs the high-temperature air from the hot channel to the air conditioning system, and at the same time delivers the cooled air to the cold channel to exchange heat and reduce the ambient temperature [10]. The water cooling system is mainly composed of a water cooling host, cooling pump, cooling tower, etc. [11], which uses water as the heat transfer medium for heat exchange and has higher cooling efficiency.

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The optimization of the data center's cooling system is basically to adjust the cooling capacity so that the temperature in the server room is within the normal working range of IT equipment [12-14]. To ensure the absolute safety of the servers in the server room, most data centers set the air conditioner temperature very low. In this way, although the normal working requirements of the server are met, there is also a lot of cooling loss, resulting in excessive consumption of electric energy, which indirectly leads to an increase in carbon emissions.

At present, in the work of achieving the goal of energy-saving in the data center through the optimization of the cooling system, many are still achieved through the experience of traditional manual experts. This implementation method has these disadvantages: the regulating frequency is relatively low; Large demand for labor; energy-saving effect is limited. In some work, the data center energy-saving system with cloud computing architecture is used, and AI is used to replace expert experience to formulate adjustment strategies. In these works, the intelligent modules are concentrated on the cloud, and the environment information of the server room is used as input to send requests to the cloud to obtain the adjustment strategy. This method also has some limits such as long network link and high network delay. The node failure in the network will cause the failure of calling the intelligent module service. This paper proposes an energy-saving system for Data Center based on fog computing architecture. The intelligent module is placed in the fog node close to the air conditioning and cooling system, which shortens the link to call AI services, improves real-time performance, and enhances stability. The main contributions of this paper are as follows:

(1) The solution of data center energy-saving system based on fog computing is to reduce the energy consumption of data center cooling system.

(2)The proposed solution can obtain the real-time data inside the server room in the data through the dynamic loop system, and use AI technology to derive energy-saving strategies through fog computing. In the energy-saving system, the AI energy-saving strategy is sent down to the end air through the dynamic loop system, thus reducing the energy consumption of the data center.

(3)The experimental results obtained from the A02-206 network room of the Xi'an New Area IDC data center in Xi'an, Shaanxi Province, show that the proposed energy-saving solution reduces the pue from 1.1929 to 1.04. And the speed of energy-saving policy generation using the fog computing based architecture is 29.84% higher than that using the cloud computing architecture.

II. Related Works

In order to ensure the safe operation of IT equipment in the data center server room, a large amount of heat generated by the IT equipment in the server room needs to be released into the environment [15]. Most of the thermal management of data centers suffers from wasted cooling.In 2012, Cisco first proposed the architecture of fog computing to make up for the lack of cloud computing [16]. Fog computing architecture, which places some computing tasks, storage and network resources close to the edge, is a highly distributed platform. This architecture is suitable for users with latency-sensitive service requirements and therefore is also growing rapidly.

IT energy-saving is the main way in data center energy-saving methods, where for every 1W of CPU power saving, the server can save 1.5W of power consumption and the data center reduces 3W of overall power consumption, Dynamic Virtual Machine Consolidation (DVMC) method dynamically optimizes the resource matching relationship between physical and virtual machines in IT equipment and is a recognized method to optimize data center energy consumption.

Data center is mainly IT equipment energy consumption and cooling equipment energy consumption.Therefore, cooling equipment also has a lot of space for energy-saving.

Traditional energy-saving methods are mostly based on expert experience in manual regulation (such as increasing and decreasing the temperature of terminal air conditioners, closing cold and hot channels, and switching on and off water cooling units) and server room equipment transformation. For new data centers, selecting frequency conversion equipment is a very good choice, but for old data centers, the cost of frequency conversion transformation is relatively high. For such data centers, it is the best choice to use AI technology to achieve energy conservation through software.

In order to reduce regulation frequency and serious loss of cooling capacity of terminal air conditioners in traditional manual regulation based on expert experience, we designed a data center energy-saving application system based on fog computing architecture.

III. System Composition

The whole system is mainly composed of a Supervision system, energy-saving system, big data lake, and cloud system. The system components is shown in figure1.

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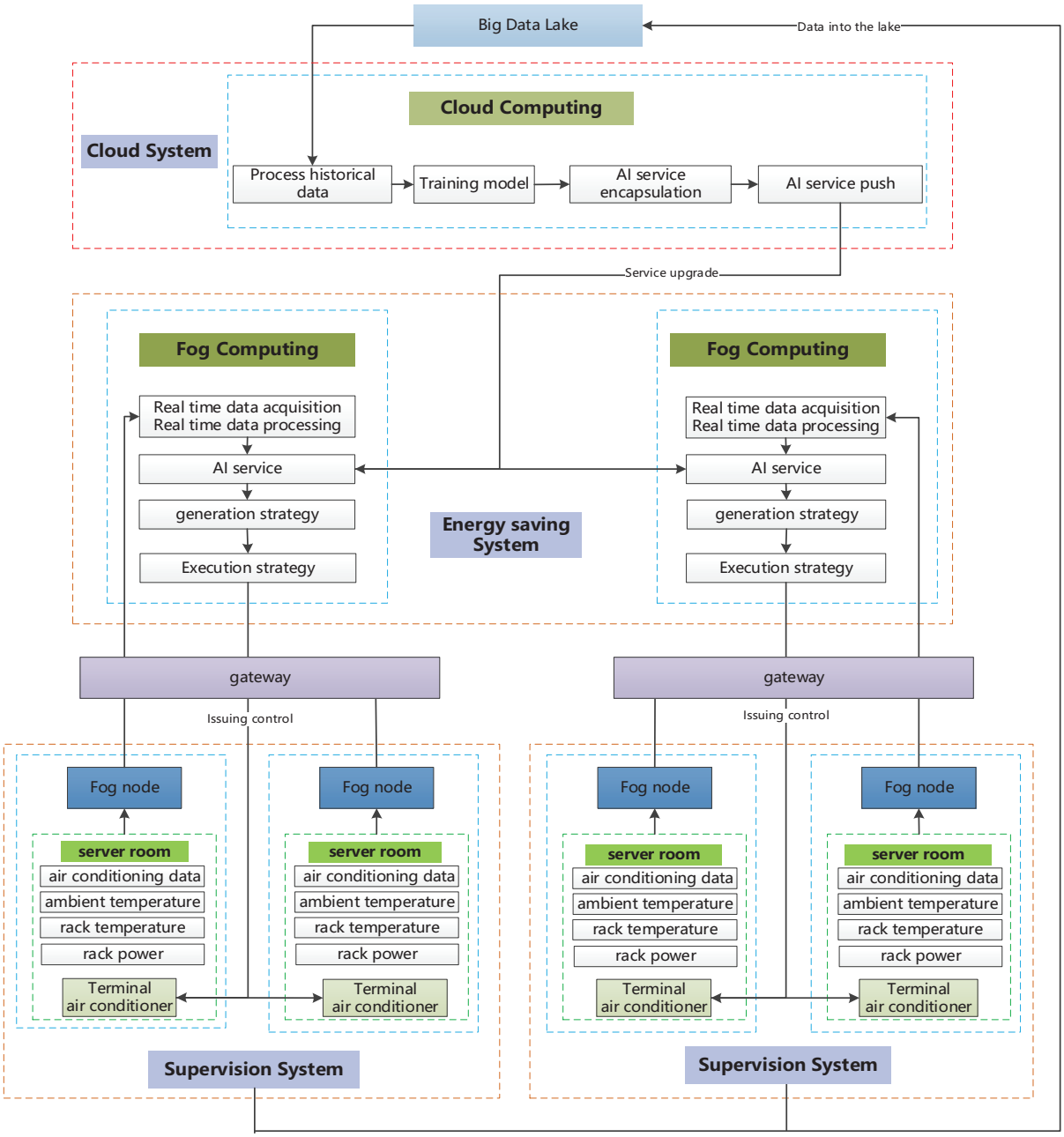

Figure 1. System architecture diagram

A. *Supervision System*

Each province has a Supervision system, each Supervision system has multiple data centers, each data center has multiple server rooms underneath, and each server room is a fog node.

The data collected by the Supervision system are mainly ambient temperature, cabinet temperature (including cabinet front (including cold channel) temperature, cabinet back door (including hot pain but) temperature), equipment load (cabinet load, air conditioning load, column head cabinet load), air conditioning data (set temperature, set humidity, air conditioning switch state, air conditioning fan speed percentage, fan rated power, fan minimum power, water valve opening set value, air conditioning return (air return temperature).

The Supervision system collects real-time data from the end equipment of the server room, saves the data for each hour, and uploads the data for each day to the group's big data lake at the end of each day. The API interface is provided to the energy-saving system so that the energy-saving system can obtain the real-time data of the end equipment.

The Issuing control can remotely regulate the adjustable parameters of the end air conditioners. The end

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air conditioner opens the regulation protocol to the Supervision system so that the Supervision system can regulate the air conditioner. The Supervision system provides an open API interface to the energy-saving system so that the energy-saving system can access the Supervision system through the API interface, and then regulate the end air conditioner.

B. Cloud System

The cloud system can obtain the historical data of the end equipment, and train the model with cloud computing on the historical data for each server room to get the AI service of a certain server room. And this service is pushed to the target energy-saving system.

C. Energy-saving System

There is a fog computing server deployed in each province to deploy a set of energy-saving system. The energy-saving strategies based on real-time data, call AI services, and regulate the end air conditioners through the Supervision system.

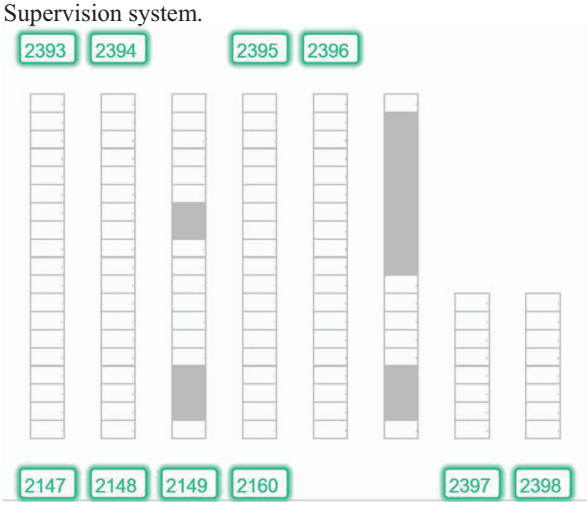

Figure 2. Server room layout

According to the API interface provided by the Supervision system, the energy-saving system obtains the adjustable parameters of the air conditioner as parameters. In turn, it generates energy-saving strategies.

After passing the audit, the strategy is issued through the Supervision system, and the end air conditioner is regulated to the strategy adjustment value, thus realizing energy-saving and consumption reduction.

D. Big Data Lake

The big data lake stores the historical data of the end equipment of the server room in the data center. It mainly includes air conditioning data, load data, cabinet temperature data, and daily power consumption

IV. Performance Evaluation

The A02-206 network server room of the Xi'an New Area IDC data center in Xi'an, Shaanxi Province. There are 10 water-cooled terminal air conditioners and 113 cabinets in this room. The layout is shown in figure2. The same type of air conditioner is used in the room, with a rated fan power consumption of 6.9KW, a rated apparent cooling capacity of 109.6kw, and a rated air volume of 26000 m^3/h. The adjustable parameters of the air conditioner are air conditioner switching status, air conditioner temperature setting value, minimum fan speed power, and maximum fan speed power.

A. Comparison with traditional manual regulation.

PUE is a metric to evaluate the energy efficiency of a data center, and the closer the rate is to 1, the more energy efficient it is.PUE can be expressed as:

$$\mathrm{PUE} = \frac{E_{total}}{E_{it}} \approx \frac{E_{it}+E_{air}}{E_{it}} \tag{1}$$

E_{total} is the total energy consumption, mainly including IT equipment energy consumption, power supply and distribution system energy consumption, cooling system energy consumption, lighting, and other energy consumption. It is approximately equal to the sum of IT equipment energy consumption E_{it} and air conditioning energy consumption and cooling system energy consumption E_{air}.

From September 6, 2022, to September 19, 2022, a total of 14 days of IT equipment energy consumption and air conditioning energy consumption data are obtained, and the data are shown in Table 1. The AI service was called from September 13, the energy saving strategy was generated and executed successfully, and the high temperature alarm of the machine room was not triggered. Specific air conditioning regulation strategies are shown in the Table 2.

TABLE I. IT EQUIPMENT LOAD AND AIR CONDITIONING ENERGY CONSUMPTION

DATE	Eit(KWH)	Eair(KWH)	PUE
09-06	4240.7	698.4	1.16
09-07	4251.4	731.1	1.17
09-08	4242.6	777.6	1.18
09-09	4244.9	827.4	1.19
09-10	4251.6	884.5	1.21
09-11	4254.4	920	1.22
09-12	4253.7	920	1.22
09-13	4249.2	173	1.04
09-14	4243.9	184	1.04
09-15	4213.4	190.13	1.05
09-16	4240.5	190.5	1.04
09-17	4220.1	184.9	1.04
09-18	4224.5	182.9	1.04
09-19	4235.3	143.25	1.04

TABLE II. STRATEGY PARAMETERS OF AIR CONDITIONING

Air_id	Min power of fan speed	Max power of fan speed
2395	from 60% to 40%	from 100% to 60%
2419	from 59% to 40%	from 80% to 60%
2160		from 90% to 80%
2397		from 80% to 100%

The trend of PUE is shown in Figure 3. It can be seen from table1 and figure3 that the air conditioning energy consumption is reduced and the PUE is reduced after the

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AI service is invoked on September 13. It shows that AI energy-saving is more energy-saving than traditional manual regulation.

B. Comparison with cloud computing

If we place AI services in the cloud and use cloud computing, we need to call the AI services in the cloud after obtaining real-time data when generating AI energy-saving policies, compared to placing AI services in provincial servers and using fog computing. The access speed of fog computing is faster and has smaller network latency.

For the experimental server room, 200 times of cloud computing to generate AI energy-saving policy and fog computing to generate AI energy-saving policy are conducted respectively, and the generation time of the policy is shown in the figure4. From the figure, it can be seen that the time to generate policy using fog computing is significantly less than the time to generate policy using cloud computing. The average time to generate AI energy-saving policy using cloud computing makes 2225.76 ms, and the average time to generate AI energy-saving policy using fog computing is 3172.6 ms. The speedup rate is (3172.6-2225.76)/2225.76*100%=29.84%.

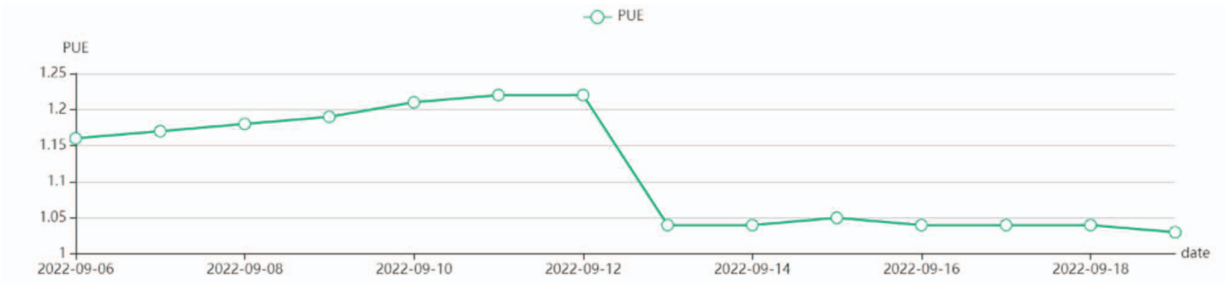

Figure 3. PUE change graph

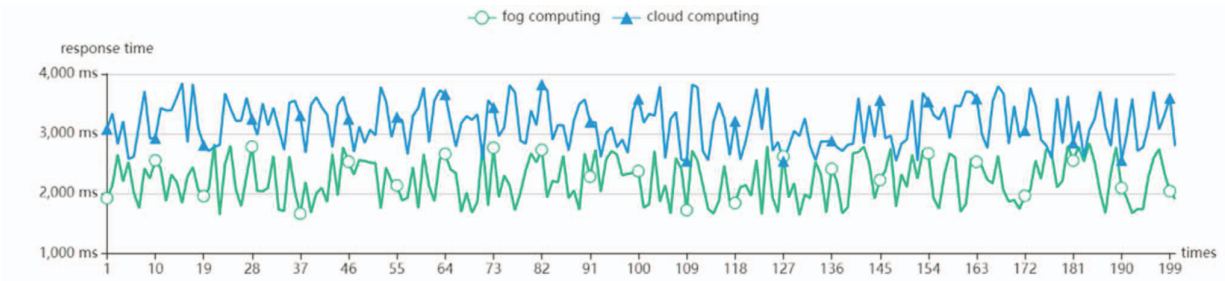

Figure 4. Comparison of cloud computing and fog computing response time

V. CONCLUSION

This paper proposes a data center energy-saving application that has been based on fog computing. The whole application includes big data lake, cloud system, energy-saving system, and Supervision system. The big data lake keeps the historical data of the end equipment in the server room, and the cloud system trains AI models based on the historical data and pushes the AI models to the energy-saving system. The energy-saving system gets real-time data from the Supervision system and invokes AI service to generate energy-saving strategies. The experimental results show that compared with the traditional expert experience manual regulation, the automatic control using the AI energy-saving strategy of the energy-saving application described in this question PUE variation graph.can achieve a PUE of 1.04 and greatly reduce the energy consumption of the data center. Compared with cloud computing for generating AI energy-saving policies, the speed of generating AI energy-saving policies using fog computing is increased by 29.84%. Therefore, data center energy-saving applications using fog computing should be promoted on a large scale.

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